Experiment 11: Write a Program to Implement Decision Tree Algorithm on Diabetes Dataset of Sklearn

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_diabetes
diabetes = load_diabetes()
```

In [2]:
```
diabetes.feature_names
```

Out[2]:
```
['age', 'sex', 'bmi', 'bp', 's1', 's2', 's3', 's4', 's5', 's6']
```

In [3]:
```
diabetes.DESCR
```

Out[3]:
```
'.. _diabetes_dataset:\n\nDiabetes dataset\n----------------\n\nTen baseline var
iables, age, sex, body mass index, average blood\npressure, and six blood serum
measurements were obtained for each of n =\n442 diabetes patients, as well as th
e response of interest, a\nquantitative measure of disease progression one year
after baseline.\n\n**Data Set Characteristics:**\n\n:Number of Instances: 442\n
\n:Number of Attributes: First 10 columns are numeric predictive values\n\n:Targ
et: Column 11 is a quantitative measure of disease progression one year after ba
seline\n\n:Attribute Information:\n    - age     age in years\n    - sex\n    -
bmi     body mass index\n    - bp      average blood pressure\n    - s1      tc,
total serum cholesterol\n    - s2      ldl, low-density lipoproteins\n    - s3
hdl, high-density lipoproteins\n    - s4      tch, total cholesterol / HDL\n
- s5      ltg, possibly log of serum triglycerides level\n    - s6      glu, blo
od sugar level\n\nNote: Each of these 10 feature variables have been mean center
ed and scaled by the standard deviation times the square root of `n_samples` (i.
e. the sum of squares of each column totals 1).\n\nSource URL:\nhttps://www4.sta
t.ncsu.edu/~boos/var.select/diabetes.html\n\nFor more information see:\nBradley
Efron, Trevor Hastie, Iain Johnstone and Robert Tibshirani (2004) "Least Angle R
egression," Annals of Statistics (with discussion), 407-499.\n(https://web.stanf
ord.edu/~hastie/Papers/LARS/LeastAngle_2002.pdf)\n'
```

In [4]:
```
df = pd.DataFrame(diabetes.data , columns= diabetes.feature_names)
df.head()
```

Out[4]:

	age	sex	bmi	bp	s1	s2	s3	s4
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592

In [5]:
```
df['target']= diabetes.target
df.head()
```

Out[5]:

	age	sex	bmi	bp	s1	s2	s3	s4
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592

In [6]:
```
from sklearn.model_selection import train_test_split
```

In [7]:
```
x = df.iloc[:,:-1]
y= df.iloc[:,-1]
```

In [8]:
```
x_train , x_test , y_train , y_test = train_test_split(x,y,test_size= 0.2)
```

In [9]:
```
from sklearn import tree
model = tree.DecisionTreeRegressor()
```

In [10]:
```
model.fit(x_train , y_train)
```

Out[10]:

▾ DecisionTreeRegressor

▸ Parameters

▸ Fitted attributes

In [11]:
```
model.score(x_train , y_train)
```

Out[11]: 1.0